



**Generation Interconnection**  
**REVISED**  
**System Impact Study**  
**Report for**  
**Queue Project AF1-083**  
**GREEN COUNTY-SALOMA 161 KV**  
**33 MW Capacity / 55 MW Energy**

July, 2022

## Table of Contents

|        |                                                              |    |
|--------|--------------------------------------------------------------|----|
| 1      | Introduction.....                                            | 4  |
| 2      | Preface.....                                                 | 4  |
| 3      | General .....                                                | 5  |
| 4      | Point of Interconnection.....                                | 6  |
| 5      | Cost Summary .....                                           | 6  |
| 6      | Transmission Owner Scope of Work .....                       | 7  |
| 6.1    | Attachment Facilities.....                                   | 7  |
| 6.2    | Direct Connection Cost Estimate .....                        | 7  |
| 6.3    | Non-Direct Connection Cost Estimate.....                     | 7  |
| 7      | Incremental Capacity Transfer Rights (ICTRs) .....           | 9  |
| 8      | Interconnection Customer Requirements.....                   | 9  |
| 9      | Revenue Metering and SCADA Requirements .....                | 10 |
| 9.1    | PJM Requirements .....                                       | 10 |
| 9.2    | Meteorological Data Reporting Requirements .....             | 10 |
| 9.3    | Interconnected Transmission Owner Requirements.....          | 10 |
| 10     | Summer Peak Analysis .....                                   | 11 |
| 10.1   | Generation Deliverability .....                              | 11 |
| 10.2   | Multiple Facility Contingency .....                          | 11 |
| 10.3   | Contribution to Previously Identified Overloads.....         | 11 |
| 10.4   | Steady-State Voltage Requirements.....                       | 12 |
| 10.5   | Potential Congestion due to Local Energy Deliverability..... | 12 |
| 10.6   | System Reinforcements .....                                  | 13 |
| 10.7   | Flow Gate Details .....                                      | 14 |
| 10.7.1 | Index 1 .....                                                | 15 |
| 10.7.2 | Index 2 .....                                                | 15 |
| 10.7.3 | Index 3.....                                                 | 15 |
| 10.7.4 | Index 4.....                                                 | 16 |
| 10.7.5 | Index 5.....                                                 | 17 |
| 10.7.6 | Index 6.....                                                 | 18 |
| 10.8   | Queue Dependencies .....                                     | 19 |
| 10.9   | Contingency Descriptions .....                               | 20 |

11 Light Load Analysis .....22

11.1 Generation Deliverability .....22

11.2 Multiple Facility Contingency .....22

11.3 Contribution to Previously Identified Overloads.....22

11.4 Steady-State Voltage Requirements .....22

11.5 Potential Congestion due to Local Energy Deliverability .....22

11.6 System Reinforcements.....23

12 Short Circuit Analysis .....24

13 Stability and Reactive Power Requirements for Low Voltage Ride Through .....24

14 Affected Systems.....24

14.1 TVA .....24

14.2 Duke Energy Progress .....24

14.3 MISO.....24

14.4 LG&E.....24

15 Attachment 1: One- Line Diagram.....25

16 Attachment 2: Stability Analysis Results.....26

## 1 Introduction

This System Impact Study has been prepared in accordance with the PJM Open Access Transmission Tariff, 205, as well as the System Impact Study Agreement between the Interconnection Customer (IC), and PJM Interconnection, LLC (PJM), Transmission Provider (TP). The Interconnected Transmission Owner (ITO) is EKPC.

## 2 Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

The Interconnection Customer seeking to interconnect a wind or solar generation facility shall maintain meteorological data facilities as well as provide that meteorological data which is required per Schedule H to the Interconnection Service Agreement and Section 8 of Manual 14D.

### 3 General

The Interconnection Customer (IC), has proposed a Solar; Storage generating facility located in Taylor County, West Virginia. The installed facilities will have a total capability of 55 MW with 33 MW of this output being recognized by PJM as Capacity.

The proposed in-service date for this project is December 31, 2022. This study does not imply a TO commitment to this in-service date.

The objective of this System Impact Study is to determine budgetary cost estimates and approximate construction timelines for identified transmission facilities required to connect the proposed generating facilities to the ITO transmission system. These reinforcements include the Attachment Facilities, Local Upgrades, and Network Upgrades required for maintaining the reliability of the ITO transmission system.

|                            |                            |
|----------------------------|----------------------------|
| <b>Queue Number</b>        | <b>AF1-083</b>             |
| <b>Project Name</b>        | GREEN COUNTY-SALOMA 161 KV |
| <b>State</b>               | West Virginia              |
| <b>County</b>              | Taylor                     |
| <b>Transmission Owner</b>  | EKPC                       |
| <b>MFO</b>                 | 55                         |
| <b>MWE</b>                 | 55                         |
| <b>MWC</b>                 | 33                         |
| <b>Fuel</b>                | Solar; Storage             |
| <b>Basecase Study Year</b> | 2023                       |

Any new service customers who can feasibly be commercially operable prior to June 1st of the basecase study year are required to request interim deliverability analysis.

## 4 Point of Interconnection

AF1-083 will interconnect with the EKPC transmission system tapping the Green County to Saloma 161 kV line.

## 5 Cost Summary

The AF1-083 project will be responsible for the following costs:

| Description                                     | Total Cost         |
|-------------------------------------------------|--------------------|
| Attachment Facilities                           | \$610,000          |
| Direct Connection Network Upgrade               | \$5,420,000        |
| Non Direct Connection Network Upgrades          | \$2,110,000        |
| Allocation for New System Upgrades*             | \$0                |
| Contribution to Previously Identified Upgrades* | \$0                |
| <b>Total Costs</b>                              | <b>\$8,140,000</b> |

\*As your project progresses through the study process and other projects modify their request or withdraw, then your cost allocation could change.

This cost excludes a Federal Income Tax Gross Up charges. This tax may or may not be charged based on whether this project meets the eligibility requirements of IRS Notice 88-129. If at a future date it is determined that the Federal Income Tax Gross charge is required, the Transmission Owner shall be reimbursed by the Interconnection Customer for such taxes.

Note 1: PJM Open Access Transmission Tariff (OATT) section 217.3A outline cost allocation rules. The rules are further clarified in PJM Manual 14A Attachment B. The allocation of costs for a network upgrade will start with the first Queue project to cause the need for the upgrade. Later queue projects will receive cost allocation contingent on their contribution to the violation and are allocated to the queues that have not closed less than 5 years following the execution of the first Interconnection Service Agreement which identifies the need for this upgrade.

Note 2: For customers with System Reinforcements listed: If your present cost allocation to a System Reinforcement indicates \$0, then please be aware that as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, the cost responsibilities can change and a cost allocation may be assigned to your project. In addition, although your present cost allocation to a System Reinforcement is presently \$0, your project may need this system reinforcement completed to be deliverable to the PJM system. If your project comes into service prior to completion of the system reinforcement, an interim deliverability study for your project will be required.

## 6 Transmission Owner Scope of Work

### 6.1 Attachment Facilities

The total preliminary cost estimate for the Attachment work is given in the table below. These costs do not include CIAC Tax Gross-up.

| Description                                                                                                                                                                                                                                                                                                                                                                     | Total Cost       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| Install necessary equipment (a 161 kV isolation switch structure and associated switch, plus interconnection metering, fiber-optic connection and telecommunications equipment, circuit breaker and associated switches, and relay panel) at the new North Taylor County switching station, to accept the IC generator lead line/bus (Estimated time to implement is 24 months) | \$610,000        |
| <b>Total Attachment Facility Costs</b>                                                                                                                                                                                                                                                                                                                                          | <b>\$610,000</b> |

### 6.2 Direct Connection Cost Estimate

The total preliminary cost estimate for the Direct Connection work is given in the table below. These costs do not include CIAC Tax Gross-up.

| Description                                                                                                                                                                                                                     | Total Cost         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| Construct a new 161 kV switching station (North Taylor County Switching) to facilitate connection of the IC solar generation project to the existing Green County-Saloma 161 kV line (Estimated time to implement is 24 months) | \$5,420,000        |
| <b>Total Direct Connection Facility Costs</b>                                                                                                                                                                                   | <b>\$5,420,000</b> |

### 6.3 Non-Direct Connection Cost Estimate

The total preliminary cost estimate for the Non-Direct Connection work is given in the table below. These costs do not include CIAC Tax Gross-up.

| Description                                                                                                                                                             | Total Cost |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| Construct facilities to loop the existing Green County-Saloma 161 kV line into the new North Taylor County switching station (Estimated time to implement is 24 months) | \$560,000  |

| Description                                                                                                                                                                | Total Cost         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| Modify relays and/or settings at Green County substation for the existing line to the new North Taylor County switching station (Estimated time to implement is 9 months)  | \$70,000           |
| Modify relays and/or settings at Marion County substation for the existing line to the new North Taylor County switching station (Estimated time to implement is 9 months) | \$70,000           |
| Install OPGW on the North Taylor County-Green County 161 kV line (11.5 miles) (Estimated time to implement is 16 months)                                                   | \$1,410,000        |
| <b>Total Non-Direct Connection Facility Costs</b>                                                                                                                          | <b>\$2,110,000</b> |

## 7 Incremental Capacity Transfer Rights (ICTRs)

None

## 8 Interconnection Customer Requirements

It is understood that the Interconnection Customer is responsible for all costs associated with this interconnection. The costs above are reimbursable to the Interconnected Transmission Owner. The cost of the Interconnection Customer's generating plant and the costs for the line connecting the generating plant to the Interconnected Transmission Owner's Transmission circuit are not included in this report; these are assumed to be the Interconnection Customer's responsibility.

The Generation Interconnection Agreement does not in or by itself establish a requirement for the Interconnected Transmission Owner to provide power for consumption at the developer's facilities. A separate agreement may be reached with the local utility that provides service in the area to ensure that infrastructure is in place to meet this demand and proper metering equipment is installed. It is the responsibility of the developer to contact the local service provider to determine if a local service agreement is required.

Requirement from the PJM Open Access Transmission Tariff:

1. An Interconnection Customer entering the New Services Queue on or after October 1, 2012 with a proposed new Customer Facility that has a Maximum Facility Output equal to or greater than 100 MW shall install and maintain, at its expense, phasor measurement units (PMUs). See Section 8.5.3 of Appendix 2 to the Interconnection Service Agreement as well as section 4.3 of PJM Manual 14D for additional information.
2. The Interconnection Customer may be required to install and/or pay for metering as necessary to properly track real time output of the facility as well as installing metering which shall be used for billing purposes. See Section 8 of Appendix 2 to the Interconnection Service Agreement as well as Section 4 of PJM Manual 14D for additional information.

## **9 Revenue Metering and SCADA Requirements**

### **9.1 PJM Requirements**

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Section 8 of Attachment O.

### **9.2 Meteorological Data Reporting Requirements**

The solar generation facility shall provide the Transmission Provider with site-specific meteorological data including:

- Back Panel temperature (Fahrenheit)
- Irradiance (Watts/meter<sup>2</sup>)
- Ambient air temperature (Fahrenheit) – (Accepted, not required)
- Wind speed (meters/second) – (Accepted, not required)
- Wind direction (decimal degrees from true north) – (Accepted, not required)

### **9.3 Interconnected Transmission Owner Requirements**

The IC will be required to comply with all Interconnected Transmission Owner's revenue metering requirements for generation interconnection customers located at the following link:

<http://www.pjm.com/planning/design-engineering/to-tech-standards/>

## 10 Summer Peak Analysis

The Queue Project AF1-083 was evaluated as a 55.0 MW (Capacity 33.0 MW) injection tapping the Green County to Saloma 161 kV line (specifically the Taylor County Jct – Saloma Tap 161 kV line segment) in the EKPC area. Project AF1-083 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AF1-083 was studied with a commercial probability of 100.0 %. Potential network impacts were as follows:

### 10.1 Generation Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

### 10.2 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

### 10.3 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

| ID       | FROM BUS# | FROM BUS     | kV    | FROM BUS AREA | TO BUS# | TO BUS       | kV    | TO BUS AREA | CK T ID | CONT NAME                | Type    | Rating MVA | PRE PROJE CT LOADIN G % | POST PROJE CT LOADIN G % | AC/D C | MW IMPAC T |
|----------|-----------|--------------|-------|---------------|---------|--------------|-------|-------------|---------|--------------------------|---------|------------|-------------------------|--------------------------|--------|------------|
| 40972325 | 342286    | 2SOMERSET    | 69.0  | EKPC          | 342287  | 2SOMERSET KU | 69.0  | EKPC        | 1       | EKPC_P7-1_COOP 161 DBL 2 | tower   | 115.0      | 106.59                  | 110.14                   | AC     | 4.66       |
| 40972225 | 342287    | 2SOMERSET KU | 69.0  | EKPC          | 324531  | 2FERGUSON SO | 69.0  | LGEE        | 1       | EKPC_P7-1_COOP 161 DBL 2 | tower   | 105.0      | 117.59                  | 121.74                   | AC     | 5.03       |
| 40972262 | 342718    | 5COOPER2     | 161.0 | EKPC          | 324141  | 5ELIHU       | 161.0 | LGEE        | 1       | EKPC_P7-1_LAURL 161 DBL  | tower   | 277.0      | 108.76                  | 111.6                    | AC     | 7.68       |
| 41169579 | 342718    | 5COOPER2     | 161.0 | EKPC          | 324141  | 5ELIHU       | 161.0 | LGEE        | 1       | EKPC_P1-2_LAURL DAM161   | single  | 277.0      | 101.41                  | 103.12                   | AC     | 4.62       |
| 41508731 | 342718    | 5COOPER2     | 161.0 | EKPC          | 324141  | 5ELIHU       | 161.0 | LGEE        | 1       | EKPC_P4-5_LAURL S50-1024 | breaker | 277.0      | 108.76                  | 111.61                   | AC     | 7.68       |
| 41508732 | 342718    | 5COOPER2     | 161.0 | EKPC          | 324141  | 5ELIHU       | 161.0 | LGEE        | 1       | EKPC_P4-5_LAURL S50-1014 | breaker | 277.0      | 108.45                  | 111.3                    | AC     | 7.69       |
| 41781821 | 342718    | 5COOPER2     | 161.0 | EKPC          | 324141  | 5ELIHU       | 161.0 | LGEE        | 1       | EKPC_P2-2_LAURE L CO 161 | bus     | 277.0      | 108.45                  | 111.3                    | AC     | 7.69       |

## 10.4 Steady-State Voltage Requirements

See Attachment 2.

## 10.5 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

| ID       | FROM BUS# | FROM BUS | kV    | FROM BUS AREA | TO BUS# | TO BUS | kV    | TO BUS AREA | CKT ID | CONT NAME                 | Type      | Rating MVA | PRE PROJECT LOADING % | POST PROJECT LOADING % | AC/D C | MW IMPACT |
|----------|-----------|----------|-------|---------------|---------|--------|-------|-------------|--------|---------------------------|-----------|------------|-----------------------|------------------------|--------|-----------|
| 41169577 | 342718    | 5COOPER2 | 161.0 | EKPC          | 324141  | 5ELIHU | 161.0 | LGEE        | 1      | EKPC_P1 - 2_LAUREL DAM161 | operation | 277.0      | 108.44                | 111.3                  | AC     | 7.7       |

## 10.6 System Reinforcements

| ID                                              | Idx | Facility                                          | Upgrade Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Cost      | Cost Allocated to AF1-083 | Upgrade Number |
|-------------------------------------------------|-----|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------------------|----------------|
| 40972325                                        | 4   | 2SOMERSET 69.0 kV - 2SOMERSET KU 69.0 kV Ckt 1    | Upgrade the existing 500 MCM CU bus jumpers to 750 MCM CU. 6 month time estimate. New expected SE rating after the upgrade will be 152 MVA.<br><br>This line overload is presently driven by a prior queue cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | \$250 K   | \$0                       | N6232          |
| 40972225                                        | 5   | 2SOMERSET KU 69.0 kV - 2FERGUSON SO 69.0 kV Ckt 1 | EKPC: SE rating is 152 MVA. No EKPC upgrade required.<br><br>LG&E: SE rating is 83 MVA. A LG&E affected system study will be required to determine if LG&E upgrades are required on this line. Preliminary upgrade, if determined to be required, is to replace terminal equipment at a cost estimate of \$897.613 K.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | \$0       | \$0                       | N/A            |
| 41781821,4150 8731,41169579, 40972262,4150 8732 | 6   | 5SCOOPER2 161.0 kV - 5ELIHU 161.0 kV Ckt 1        | EKPC:<br>EKPC SE rating is 298 MVA.<br>Increase the operating temperature of the 795 MCM ACSR conductor from 212F to 275F (6.7 miles). EKPC's new SE rating would be 371 MVA. Cost estimate \$660K. PJM Network Upgrade N6238.<br><br>Note: this EKPC upgrade may be dependent upon whether LG&E determines if an LG&E end upgrade is required on this line since the equipment which limits the overall line rating is LG&E equipment. This will be determined with a LG&E affected system study.<br><br>LG&E:<br>LG&E SE rating is 277 MVA. A LG&E affected system study will be required to determine if LG&E upgrades are required on this line. Preliminary upgrade, if determined to be required, is to upgrade the line conductor at a cost estimate of \$28.083 K. New LG&E expected SE rating to be 335 MVA. | \$660 K   | \$0                       | N6238          |
| Total Cost                                      |     |                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | \$860,000 | \$0                       |                |

Note : For customers with System Reinforcements listed: If your present cost allocation to a System Reinforcement indicates \$0, then please be aware that as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, the cost responsibilities can change and a cost allocation may be assigned to your project. In addition, although your present cost allocation to a System Reinforcement is presently \$0, your project may need this system reinforcement completed to be deliverable to the PJM system. If your project comes into service prior to completion of the system reinforcement, an interim deliverability study for your project will be required.

## 10.7 Flow Gate Details

The following indices contain additional information about each facility presented in the body of the report. For each index, a description of the flowgate and its contingency was included for convenience. The intent of the indices is to provide more details on which projects/generators have contributions to the flowgate in question. All New Service Queue Requests, through the end of the Queue under study, that are contributors to a flowgate will be listed in the indices. Please note that there may be contributors that are subsequently queued after the queue under study that are not listed in the indices. Although this information is not used "as is" for cost allocation purposes, it can be used to gage the impact of other projects/generators. It should be noted the project/generator MW contributions presented in the body of the report are Full MW Impact contributions which are also noted in the indices column named "Full MW Impact", whereas the loading percentages reported in the body of the report, take into consideration the PJM Generator Deliverability Test rules such as commercial probability of each project as well as the ramping impact of "Adder" contributions. The MW Impact found and used in the analysis is shown in the indices column named "Gendeliv MW Impact".

#### 10.7.1 Index 1

None.

#### 10.7.2 Index 2

None.

#### 10.7.3 Index 3

None.

#### 10.7.4 Index 4

| ID       | FROM BUS# | FROM BUS  | FROM BUS AREA | TO BUS# | TO BUS       | TO BUS AREA | CKT ID | CONT NAME                      | Type  | Rating MVA | PRE PROJECT LOADING % | POST PROJECT LOADING % | AC DC | MW IMPACT |
|----------|-----------|-----------|---------------|---------|--------------|-------------|--------|--------------------------------|-------|------------|-----------------------|------------------------|-------|-----------|
| 40972325 | 342286    | 2SOMERSET | EKPC          | 342287  | 2SOMERSET KU | EKPC        | 1      | EKPC_P7-1_COOP<br>161 DBL<br>2 | tower | 115.0      | 106.59                | 110.14                 | AC    | 4.66      |

| Bus #       | Bus          | Gendeliv MW Impact | Type          | Full MW Impact |
|-------------|--------------|--------------------|---------------|----------------|
| 342900      | 1COOPER1 G   | 5.6852             | 50/50         | 5.6852         |
| 342903      | 1COOPER2 G   | 11.0266            | 50/50         | 11.0266        |
| 939131      | AE1-143 C    | 5.3386             | Adder         | 6.28           |
| 939132      | AE1-143 E    | 2.6443             | Adder         | 3.11           |
| 940041      | AE1-246 C O1 | 4.2405             | Adder         | 4.99           |
| 940042      | AE1-246 E O1 | 2.0651             | Adder         | 2.43           |
| 940831      | AE2-071 C    | 1.2982             | Adder         | 1.53           |
| 940832      | AE2-071 E    | 0.8655             | Adder         | 1.02           |
| 943701      | AF1-038 C    | 6.1949             | 50/50         | 6.1949         |
| 943702      | AF1-038 E    | 4.1299             | 50/50         | 4.1299         |
| 943821      | AF1-050 C    | 2.2451             | Adder         | 2.64           |
| 943822      | AF1-050 E    | 1.4967             | Adder         | 1.76           |
| 944151      | AF1-083 C O1 | 2.3786             | Adder         | 2.8            |
| 944152      | AF1-083 E O1 | 1.5858             | Adder         | 1.87           |
| 944511      | AF1-116 C    | 5.9872             | Adder         | 7.04           |
| 944512      | AF1-116 E    | 3.9915             | Adder         | 4.7            |
| 945381      | AF1-203 C    | 0.7418             | Adder         | 0.87           |
| 945382      | AF1-203 E    | 0.4946             | Adder         | 0.58           |
| WEC         | WEC          | 0.0482             | Confirmed LTF | 0.0482         |
| LGEE        | LGEE         | 0.0131             | Confirmed LTF | 0.0131         |
| CPL         | CPL          | 0.0318             | Confirmed LTF | 0.0318         |
| LGE-0012019 | LGE-0012019  | 5.0391             | LTF           | 5.0391         |
| CBM-W2      | CBM-W2       | 3.5872             | Confirmed LTF | 3.5872         |
| NY          | NY           | 0.0426             | Confirmed LTF | 0.0426         |
| TVA         | TVA          | 1.0724             | Confirmed LTF | 1.0724         |
| O-066       | O-066        | 0.5040             | Confirmed LTF | 0.5040         |
| CBM-S2      | CBM-S2       | 0.6069             | Confirmed LTF | 0.6069         |
| CBM-S1      | CBM-S1       | 5.3506             | Confirmed LTF | 5.3506         |
| G-007       | G-007        | 0.0780             | Confirmed LTF | 0.0780         |
| MADISON     | MADISON      | 0.7560             | Confirmed LTF | 0.7560         |
| MEC         | MEC          | 0.4513             | Confirmed LTF | 0.4513         |
| CBM-W1      | CBM-W1       | 1.7014             | Confirmed LTF | 1.7014         |

## 10.7.5 Index 5

| ID       | FROM BUS# | FROM BUS     | FROM BUS AREA | TO BUS# | TO BUS       | TO BUS AREA | CKT ID | CONT NAME                | Type  | Rating MVA | PRE PROJECT LOADING % | POST PROJECT LOADING % | AC DC | MW IMPACT |
|----------|-----------|--------------|---------------|---------|--------------|-------------|--------|--------------------------|-------|------------|-----------------------|------------------------|-------|-----------|
| 40972225 | 342287    | 2SOMERSET KU | EKPC          | 324531  | 2FERGUSON SO | LGEE        | 1      | EKPC_P7-1_COOP 161 DBL 2 | tower | 105.0      | 117.59                | 121.74                 | AC    | 5.03      |

| Bus #       | Bus          | Gendeliv MW Impact | Type          | Full MW Impact |
|-------------|--------------|--------------------|---------------|----------------|
| 342900      | 1COOPER1 G   | 5.5640             | 50/50         | 5.5640         |
| 342903      | 1COOPER2 G   | 10.7915            | 50/50         | 10.7915        |
| 939131      | AE1-143 C    | 6.4733             | 50/50         | 6.4733         |
| 939132      | AE1-143 E    | 3.2064             | 50/50         | 3.2064         |
| 940041      | AE1-246 C O1 | 5.3436             | Adder         | 6.29           |
| 940042      | AE1-246 E O1 | 2.6022             | Adder         | 3.06           |
| 940831      | AE2-071 C    | 1.6422             | Adder         | 1.93           |
| 940832      | AE2-071 E    | 1.0948             | Adder         | 1.29           |
| 943701      | AF1-038 C    | 8.4539             | 50/50         | 8.4539         |
| 943702      | AF1-038 E    | 5.6359             | 50/50         | 5.6359         |
| 943821      | AF1-050 C    | 2.5937             | Adder         | 3.05           |
| 943822      | AF1-050 E    | 1.7291             | Adder         | 2.03           |
| 944151      | AF1-083 C O1 | 2.5632             | Adder         | 3.02           |
| 944152      | AF1-083 E O1 | 1.7088             | Adder         | 2.01           |
| 944511      | AF1-116 C    | 7.2598             | 50/50         | 7.2598         |
| 944512      | AF1-116 E    | 4.8398             | 50/50         | 4.8398         |
| 945381      | AF1-203 C    | 0.9384             | Adder         | 1.1            |
| 945382      | AF1-203 E    | 0.6256             | Adder         | 0.74           |
| WEC         | WEC          | 0.0621             | Confirmed LTF | 0.0621         |
| CPLE        | CPLE         | 0.0648             | Confirmed LTF | 0.0648         |
| LGE-0012019 | LGE-0012019  | 5.1436             | LTF           | 5.1436         |
| CBM-W2      | CBM-W2       | 4.6437             | Confirmed LTF | 4.6437         |
| NY          | NY           | 0.0431             | Confirmed LTF | 0.0431         |
| TVA         | TVA          | 1.4154             | Confirmed LTF | 1.4154         |
| O-066       | O-066        | 0.5107             | Confirmed LTF | 0.5107         |
| CBM-S2      | CBM-S2       | 1.0115             | Confirmed LTF | 1.0115         |
| CBM-S1      | CBM-S1       | 6.9864             | Confirmed LTF | 6.9864         |
| G-007       | G-007        | 0.0790             | Confirmed LTF | 0.0790         |
| MADISON     | MADISON      | 0.9959             | Confirmed LTF | 0.9959         |
| MEC         | MEC          | 0.5816             | Confirmed LTF | 0.5816         |
| CBM-W1      | CBM-W1       | 2.2143             | Confirmed LTF | 2.2143         |

## 10.7.6 Index 6

| ID       | FROM BUS# | FROM BUS | FROM BUS AREA | TO BUS# | TO BUS | TO BUS AREA | CKT ID | CONT NAME               | Type | Rating MVA | PRE PROJECT LOADING % | POST PROJECT LOADING % | AC DC | MW IMPACT |
|----------|-----------|----------|---------------|---------|--------|-------------|--------|-------------------------|------|------------|-----------------------|------------------------|-------|-----------|
| 41781821 | 342718    | 5COOPER2 | EKPC          | 324141  | SELIHU | LGEE        | 1      | EKPC_P2-2_LAUREL CO 161 | bus  | 277.0      | 108.45                | 111.3                  | AC    | 7.69      |

| Bus #       | Bus          | Gendeliv MW Impact | Type          | Full MW Impact |
|-------------|--------------|--------------------|---------------|----------------|
| 342900      | 1COOPER1 G   | 11.4249            | 50/50         | 11.4249        |
| 342903      | 1COOPER2 G   | 22.2261            | 50/50         | 22.2261        |
| 342945      | 1LAUREL 1G   | 6.9148             | 50/50         | 6.9148         |
| 939131      | AE1-143 C    | 10.1012            | 50/50         | 10.1012        |
| 939132      | AE1-143 E    | 5.0034             | 50/50         | 5.0034         |
| 940041      | AE1-246 C O1 | 9.1151             | 50/50         | 9.1151         |
| 940042      | AE1-246 E O1 | 4.4389             | 50/50         | 4.4389         |
| 940831      | AE2-071 C    | 2.5805             | 50/50         | 2.5805         |
| 940832      | AE2-071 E    | 1.7203             | 50/50         | 1.7203         |
| 943701      | AF1-038 C    | 6.7007             | 50/50         | 6.7007         |
| 943702      | AF1-038 E    | 4.4671             | 50/50         | 4.4671         |
| 943821      | AF1-050 C    | 4.5587             | 50/50         | 4.5587         |
| 943822      | AF1-050 E    | 3.0391             | 50/50         | 3.0391         |
| 944151      | AF1-083 C O1 | 4.6160             | 50/50         | 4.6160         |
| 944152      | AF1-083 E O1 | 3.0774             | 50/50         | 3.0774         |
| 944511      | AF1-116 C    | 11.3285            | 50/50         | 11.3285        |
| 944512      | AF1-116 E    | 7.5523             | 50/50         | 7.5523         |
| 945381      | AF1-203 C    | 1.4746             | 50/50         | 1.4746         |
| 945382      | AF1-203 E    | 0.9830             | 50/50         | 0.9830         |
| WEC         | WEC          | 0.0740             | Confirmed LTF | 0.0740         |
| CPL         | CPL          | 0.0939             | Confirmed LTF | 0.0939         |
| LGE-0012019 | LGE-0012019  | 7.9596             | LTF           | 7.9596         |
| CBM-W2      | CBM-W2       | 6.6667             | Confirmed LTF | 6.6667         |
| NY          | NY           | 0.0874             | Confirmed LTF | 0.0874         |
| TVA         | TVA          | 2.1266             | Confirmed LTF | 2.1266         |
| O-066       | O-066        | 1.0282             | Confirmed LTF | 1.0282         |
| CBM-S2      | CBM-S2       | 1.5375             | Confirmed LTF | 1.5375         |
| CBM-S1      | CBM-S1       | 10.3348            | Confirmed LTF | 10.3348        |
| G-007       | G-007        | 0.1591             | Confirmed LTF | 0.1591         |
| MADISON     | MADISON      | 1.5886             | Confirmed LTF | 1.5886         |
| MEC         | MEC          | 0.7913             | Confirmed LTF | 0.7913         |
| CBM-W1      | CBM-W1       | 2.5771             | Confirmed LTF | 2.5771         |

## 10.8 Queue Dependencies

The Queue Projects below are listed in one or more indices for the overloads identified in your report. These projects contribute to the loading of the overloaded facilities identified in your report. The percent overload of a facility and cost allocation you may have towards a particular reinforcement could vary depending on the action of these earlier projects. The status of each project at the time of the analysis is presented in the table. This list may change as earlier projects withdraw or modify their requests.

| Queue Number | Project Name                         | Status |
|--------------|--------------------------------------|--------|
| AE1-143      | Marion County 161 kV                 | Active |
| AE1-246      | Barren County-Summer Shade 161 kV    | Active |
| AE2-071      | Patton Rd-Summer Shade 69 kV         | Active |
| AF1-038      | Sewellton Jct-Webbs Crossroads 69 kV | Active |
| AF1-050      | Summer Shade - Green County 161 kV   | Active |
| AF1-083      | Green County-Saloma 161 kV           | Active |
| AF1-116      | Marion County 161 kV                 | Active |
| AF1-203      | Patton Rd-Summer Shade 69 kV         | Active |

## 10.9 Contingency Descriptions

| Contingency Name                | Contingency Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>EKPC_P2-3_MAR W38-1014</b>   | CONTINGENCY 'EKPC_P2-3_MAR W38-1014' /* MARION CO<br>OPEN BRANCH FROM BUS 324280 TO BUS 342769 CKT 1 /* 324280 4MARIONL<br>138.00 342769 5MARION CO 161.00<br>OPEN BRANCH FROM BUS 342703 TO BUS 342769 CKT 1 /* 342703 5CASEY CO<br>161.00 342769 5MARION CO 161.00<br>OPEN BRANCH FROM BUS 341269 TO BUS 342703 CKT 1 /* 341269 2CASEY CO<br>69.000 342703 5CASEY CO 161.00<br>OPEN BRANCH FROM BUS 342703 TO BUS 342760 CKT 1 /* 342703 5CASEY CO<br>161.00 342760 5LIBERTY J 161.00<br>END |
| <b>EKPC_P4-2_MAR W38-1014</b>   | CONTINGENCY 'EKPC_P4-2_MAR W38-1014' /* MARION CO<br>OPEN BRANCH FROM BUS 324280 TO BUS 342769 CKT 1 /* 324280 4MARIONL<br>138.00 342769 5MARION CO 161.00<br>OPEN BRANCH FROM BUS 342703 TO BUS 342769 CKT 1 /* 342703 5CASEY CO<br>161.00 342769 5MARION CO 161.00<br>OPEN BRANCH FROM BUS 341269 TO BUS 342703 CKT 1 /* 341269 2CASEY CO<br>69.000 342703 5CASEY CO 161.00<br>OPEN BRANCH FROM BUS 342703 TO BUS 342760 CKT 1 /* 342703 5CASEY CO<br>161.00 342760 5LIBERTY J 161.00<br>END |
| <b>EKPC_P1-2_LAUR-L DAM161</b>  | CONTINGENCY 'EKPC_P1-2_LAUR-L DAM161' /* LAUREL CO - LAUREL DAM<br>OPEN BRANCH FROM BUS 342754 TO BUS 342757 CKT 1 /* 342754 5LAUREL CO<br>161.00 342757 5LAUREL DAM 161.00<br>END                                                                                                                                                                                                                                                                                                             |
| <b>EKPC_P2-2_LAUREL CO 161</b>  | CONTINGENCY 'EKPC_P2-2_LAUREL CO 161' /* LAUREL 161 BUS<br>OPEN BUS 342754 /* 5LAUREL CO<br>END                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>EKPC_P1-2_COOP-ELIHU161</b>  | CONTINGENCY 'EKPC_P1-2_COOP-ELIHU161' /* COOPER - KU ELIHU<br>OPEN BRANCH FROM BUS 324141 TO BUS 342718 CKT 1 /* 324141 5ELIHU 161.00<br>342718 5COOPER2 161.00<br>END                                                                                                                                                                                                                                                                                                                         |
| <b>EKPC_P7-1_COOP 161 DBL 2</b> | CONTINGENCY 'EKPC_P7-1_COOP 161 DBL 2' /* COOPER - ELIHU 161 &<br>COOPER - LAUREL DAM 161<br>OPEN BRANCH FROM BUS 324141 TO BUS 342718 CKT 1 /* 324141 5ELIHU 161.00<br>342718 5COOPER2 161.00<br>OPEN BRANCH FROM BUS 342718 TO BUS 342757 CKT 1 /* 342718 5COOPER2<br>161.00 342757 5LAUREL DAM 161.00<br>END                                                                                                                                                                                |

| Contingency Name                | Contingency Definition                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>EKPC_P7-1_LAURL 161 DBL</b>  | CONTINGENCY 'EKPC_P7-1_LAURL 161 DBL' /* LAUREL CO - LAUREL DAM 161<br>& LAUREL CO - TYNER 161<br>OPEN BRANCH FROM BUS 342754 TO BUS 342757 CKT 1 /* 342754 5LAUREL CO<br>161.00 342757 5LAUREL DAM 161.00<br>OPEN BRANCH FROM BUS 342754 TO BUS 342781 CKT 1 /* 342754 5LAUREL CO<br>161.00 342781 5PITTSBURG 161.00<br>OPEN BRANCH FROM BUS 342781 TO BUS 342820 CKT 1 /* 342781 5PITTSBURG<br>161.00 342820 5TYNER 161.00<br>END |
| <b>EKPC_P4-5_LAURL S50-1014</b> | CONTINGENCY 'EKPC_P4-5_LAURL S50-1014' /* LAUREL CO<br>OPEN BUS 342754 /* 5LAUREL CO DROPS BUS<br>END                                                                                                                                                                                                                                                                                                                               |
| <b>Base Case</b>                |                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>EKPC_P4-5_LAURL S50-1024</b> | CONTINGENCY 'EKPC_P4-5_LAURL S50-1024' /* LAUREL CO<br>OPEN BUS 342754 /* 5LAUREL CO DROPS BUS<br>OPEN BRANCH FROM BUS 324688 TO BUS 342781 CKT 1 /* 324688 2PITTSKU 69.000<br>342781 5PITTSBURG 161.00<br>OPEN BRANCH FROM BUS 342781 TO BUS 342820 CKT 1 /* 342781 5PITTSBURG<br>161.00 342820 5TYNER 161.00<br>END                                                                                                               |

## 11 Light Load Analysis

The Queue Project AF1-083 was evaluated as a 26.0 MW injection/withdrawal (battery) tapping the Green County to Saloma 161 kV line (specifically the Taylor County Jct – Saloma Tap 161 kV line segment) in the EKPC area. Project AF1-083 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AF1-083 was studied with a commercial probability of 100.0 %. Potential network impacts were as follows:

### 11.1 Generation Deliverability

(Single or N-1 contingencies)

None

### 11.2 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies)

None

### 11.3 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

### 11.4 Steady-State Voltage Requirements

None

### 11.5 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

None

## 11.6 System Reinforcements

None

## 12 Short Circuit Analysis

The following Breakers are overdutied

None

## 13 Stability and Reactive Power Requirements for Low Voltage Ride Through

*(Summary of the VAR requirements based upon the results of the dynamic studies)*

See Attachment 2.

## 14 Affected Systems

### 14.1 TVA

TVA Impacts to be determined during later study phases (as applicable).

### 14.2 Duke Energy Progress

Duke Energy Progress Impacts to be determined during later study phases (as applicable).

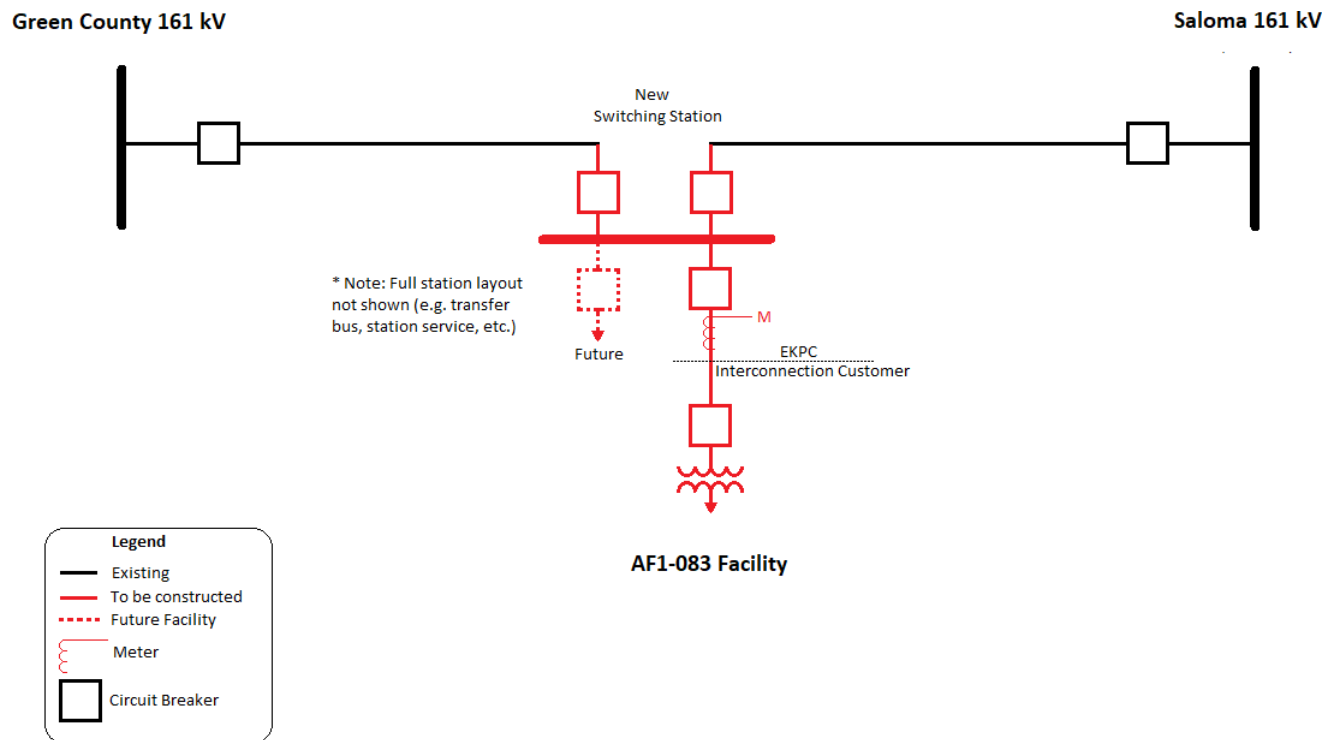
### 14.3 MISO

MISO Impacts to be determined during later study phases (as applicable).

### 14.4 LG&E

An LG&E Affected System Study will be required. PJM has identified several EKPC-LG&E tie line overloads with limiting equipment on the LG&E side. LG&E will need to determine if LG&E upgrades are required.

## 15 Attachment 1: One- Line Diagram



## 16 Attachment 2: Stability Analysis Results

### EXECUTIVE SUMMARY

Generator Interconnection Request AF1-083 for a 55 MW photovoltaic (PV) solar generation plant consisting of 13 x 4.277 MW PV SMA SC 4400 UP inverters. AF1-083 has a Point of Interconnection (POI) connecting to the transmission line between Taylor and Saloma 161 kV in Taylor, Kentucky, in the EKPC transmission system.

The power flow scenario for the analysis was based on the RTEP 2023 summer peak case, modified to include applicable queue projects. AF1-083 has been dispatched online at maximum facility output, with approximately unity power factor at the high side of the station transformer.

AF1-083 was tested for compliance with NERC, PJM, Transmission Owner and other applicable criteria. For this study, 110 contingencies were simulated, each with a 20 second simulation time period. Studied faults included:

- Steady-state operation (20 second simulation)
- Three-phase faults with normal clearing time
- Single-phase fault with normal clearing
- Single-phase faults with a stuck breaker with delayed clearing time
- Single-phase faults placed at 80% of the line with delayed (Zone 2) clearing at remote line end because of primary communications/relaying failure

The 110 fault contingencies tested on the 2023 summer peak case met the recovery criteria:

- The AF1-083 generator was able to ride through the faults except for faults where protective actions trip one or more generator(s).
- All generators maintained synchronism and any post-contingency oscillations are positively damped with a damping margin of at least 3%.
- All bus voltages recover to 0.7 p.u. within 2.5-seconds and the final voltage is within the range of 0.90 p.u. to 1.06 p.u.
- No transmission element trips, other than those either directly connected or designated to trip as a consequence of the fault.

The AF1-083 queue project did not meet the 0.95 lagging power factor requirement. An additional -15.48 Mvar would be required for the plant to meet the 0.95 lagging power factor requirement. The plant did meet the 0.95 leading power factor requirement.